

AGUSTINUS ZEFANYA

Mobile Apps Developer
Game Development and AI Interest



Personal info

- * 21/08/2003
- +62-81318463100
- agustinuszefanya@gmail.com
- Depok & North Jakarta

Links

- /catashX
- /agustinus-zefanya
- zephx.itch.io

Skills

- Flutter (Dart), C, C++, Python, Unity (C#), Swift, Javascript, Java, Kotlin, PostgreSQL, R
- LaTeX, HTML, CSS
- {MS, Libre, GSuite} Office, Adobe Acrobat
- Linux, Windows, macOS
- Visual Studio, Visual Studio Code, Anti Gravity
- Figma, Git/Github, Canva

WORK EXPERIENCES

PT Porto Indonesia Sejahtera

Intern, Mobile Apps Developer

SEPT 2024 – MARCH 2025

- Contributed to end-to-end development of internal mobile CRM applications using Flutter.
 - Built responsive UI, integrated REST APIs, tested, and deployed applications.
 - Maintained release versions and supported internal users.
- Developed and maintained Sales, Customer, and HRIS applications.
 - Added new features, fixed bugs, and improved app stability and performance.
 - Enhanced workflow efficiency based on operational needs.
- Collaborated with cross-functional teams to deliver scalable business solutions.

PT Porto Indonesia Sejahtera

Staff, Mobile Apps Developer

MARCH 2025 – NOW

- Maintained and developed internal business applications including CRM, Sales, Customer, and HRIS.
 - Implemented new features and enhanced existing modules based on business needs.
 - Improved application stability, usability, and performance.
- Contributed to cross-functional product development across mobile, backend, and native platforms.
 - Supported Portal App development and REST API services using Express.js.
 - Built native integrations and platform-specific features using Kotlin and Swift.
- Developed QC Automation System for footwear inspection using computer vision.
 - Integrated YOLO and SAM models for automated quality detection.
 - Connected sensors, conveyors, and pushers with Qt / Py-Side6 desktop interface.
- Collaborated with operations and technical teams to deliver scalable solutions.

Computer Science, IPB University
(Bachelor Degree)

INSTITUT PERTANIAN BOGOR, JAWA BARAT AUG 2021 – AUG 2025
Degree: Bachelor of Computer (S.Kom)

Grade: 3.74

Subsidiary subject: Computational Intelligence and Optimization
In-depth study: Multi-Agent, Optimization
Thesis subject: Swarm-UAV, Coordination Algorithm

SMAN 8 Depok (High School)

DEPOK 2018 – SEP 2021
Participated multiple times in National Science Olympiad (Physics). Achieved top 3 rankings several times.

PROJECTS

Courier Conquest

AGATE ACADEMY (INDEPENDENT STUDY) 2023
Developed a fantasy adventure/RPG game set in the world of Eldoria, where players complete delivery missions, explore the environment, and overcome enemies and obstacles. Contributed as Game Designer, Game Programmer, and Technical Artist, responsible for designing gameplay systems and mechanics, implementing core features, and supporting visual/technical integration to ensure smooth gameplay experience. Collaborated within a multidisciplinary team throughout the development process.

Unity, Game Design, Game Programming, HLSL Shading, C#, OOP, Behaviour tree.

Full game portfolio : [click to redirect to itch page](#)

Pengembangan dan Analisis Kinerja Algoritma Next Best View pada Sistem Multi-UAV untuk Cakupan 3D Tanaman

IPB UNIVERSITY (BACHELOR THESIS) 2024 – 2025
Developed and analyzed an optimization algorithm for multi-UAV systems to maximize 3D plant surface coverage in controlled environment agriculture. Designed a mathematical modelling and simulation framework for autonomous drone placement, sensor visibility analysis, and system performance benchmarking. Compared heuristic and metaheuristic approaches through extensive experiments to evaluate coverage efficiency, drone utilization, and computational speed.

- Built simulation environment using Python for multi-agent UAV coordination and 3D geometry processing.
- Implemented Next Best View (Greedy Optimization) and Genetic Algorithm for complex system modelling.
- Performed parameter tuning, trade-off analysis, and computational complexity evaluation.
- Modelled camera field-of-view, occlusion detection, and visibility mapping on 3D mesh plant models.
- Produced performance reports, data visualization, and comparative statistical analysis.

Skill: Python, Optimization, Simulation, Mathematical Modelling, Genetic Algorithm, UAV Systems, 3D Geometry, Data Analysis

Referenced thesis : [click to see full paper](#)

AGUSTINUS
ZEFANYA

Languages

-  Indonesian (Mothertounge)
-  English (Fluent)

Certificates and Prizes

- 2021 Participant of *7 Habits of Highly Effective People* Training
- 2022 **GameJamPlus Indonesia 2022** — Honorable Mention (*Uprising*)
- 2023 **Jogja Information Technology Session (JOINTS): Game Jam 2023** — 1st Place Best Sounds (*Puaka*)
- 2023 **Pagelaran Mahasiswa Nasional Bidang Teknologi Informasi dan Komunikasi (GEMASTIK XVI)** — Finalist (*Archemist's Chronicles*)
- 2024 Certified in Game Programmer Competency — Agate Academy Independent Study Program

Hobbies

-  Game Development
-  Swimming
-  Badminton
-  Listening to Music

Other Projects

- Porto Sales Apps
- Porto Portal Apps
- Porto HRIS Apps
- Porto Customer Apps

PROJECTS

Porto CRM

PT. PORTO INDONESIA SEJAHTERA 2024 – 2025

Developed an internal Customer Relationship Management (CRM) system to support sales operations, customer management, and business workflow automation. Led the project lifecycle from system design, UI Design, application development, backend API integration, deployment, and post-release maintenance. Built scalable features to improve operational efficiency, streamline sales processes, and centralize business data management.

Skill: Fullstack Development, Flutter, Dart, Swift, Kotlin, API Development, Database Management, System Design, Deployment, Maintenance, Business Process Automation

QC Automation System

PT. PORTO INDONESIA SEJAHTERA 2025 – 2026

Developed an intelligent Quality Control (QC) automation system using computer vision and real-time industrial sensing to improve inspection accuracy, consistency, and production efficiency. Designed the complete system architecture, operational workflow, and user interface by integrating conveyor cameras, sensors, actuators, and AI-based decision modules. Built the full software stack from frontend to backend, including machine vision processing, control logic, data monitoring, and deployment for production use. Applied analytical problem solving, system modelling, and continuous optimization to resolve recurring production issues and improve overall system reliability.

- Designed end-to-end automation method for real-time product inspection and defect classification.
- Built desktop application using PySide6 (Qt) for operator interface, machine monitoring, and control systems.
- Integrated YOLO and Segment Anything Model (SAM) for object detection, segmentation, and visual quality analysis.
- Developed backend logic for sensor communication, actuator triggering, and automated pass/fail decision systems.
- Performed data collection, statistical analysis, model validation, and parameter tuning for higher detection accuracy.
- Solved recurring operational issues through root cause analysis and process optimization.
- Improved production throughput by reducing manual inspection dependency and increasing consistency.

Skill: Python, PySide6, Qt, Computer Vision, YOLO, SAM, Automation Systems, System Modelling, Optimization, Signal Processing, Sensor Integration, Fullstack Development, Data Analysis, Problem Solving